SEAT NO. CT- 23025

NED UNIVERSITY OF ENGINEERING & TECHNOLOGY

THIRD YEAR(Computer Science)

FALL SEMESTER EXAMINATIONS 2025

Time : 3 Hours | Batch 2023 | Dated : 15-DEC-25

Max Marks : 60

Machine Learning - CT-378

Note: Attempt all questions. Answer all parts of a question in a row. State your assumptions clearly. Draw diagram where necessary.

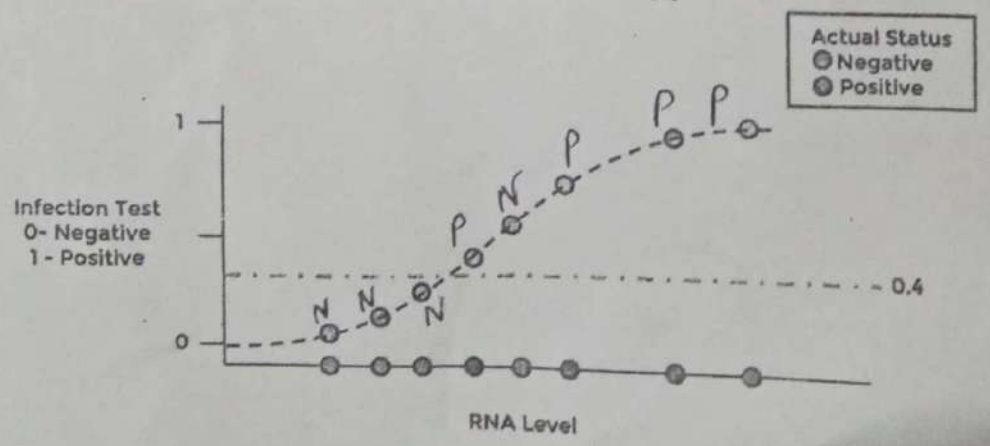
Q1.	a) ***Explain*** the zero probability problem in Naïve Bayes classifier and how to overcome this problem. Show the mathematical equation. **[4]** b) ***Explain*** briefly the methods of resolving overfitting in decision tree. Also list down the methods of measuring node impurity. **[4]** c) In a classification problem, it is possible for a classifier to face a tie situation i.e. two or more class labels receive the same prediction score or probability for a given instance. ***Describe*** the methods or strategies commonly used by classifiers to resolve a tie during prediction. **[2]** d) ***Outline*** any four commonly used regression metrics. **[2]**	CLO 1 12 Marks
Q2.	***Analyze*** appropriate Machine Learning techniques on the given dataset. a) A machine learning model fitted on a sentiment analysis dataset. ***Apply*** the following metrics to evaluate the model's performance: Confusion Matrix, Accuracy, Support, Weighted Precision, Weighted Recall, and F-Measure. Consider the following outcomes; **[6]** • **Worst:** 93 were correctly identified, 5 were misidentified as neutral, 12 were misclassified as positive • **Neutral:** 35 were correctly identified, 6 were misidentified as negative, 9 were misclassified as positive • **Positive:** 85 were correctly identified, 11 were misidentified as negative, 9 were misclassified as neutral b) A tiny dataset consist of 5 samples = $\{A, B, C, D, E\}$, using a Leave-P-Out CV approach, split the dataset such that each validation set contains exactly two samples, and the remaining samples form the training set. List all the possible combination of train-validation splits. Also determine how many times each sample appears in the validation set across all possible combinations. **[6]**	CLO 2 12 Marks
Q3.	a) ***Classify*** using Naïve Bayes Algorithm to the following table and predict class outcome of stolen for Color=Yellow, Type=Sports, and Origin=Domestic. **[6]**	CLO 2 12 Marks

Example No.	Color	Type	Origin	Stolen?
1	Red	Sports	Domestic	Yes
2	Red	Sports	Domestic	No
3	Red	Sports	Domestic	Yes
4	Yellow	Sports	Domestic	No
5	Yellow	Sports	Imported	Yes
6	Yellow	SUV	Imported	No
7	Yellow	SUV	Imported	Yes
8	Yellow	SUV	Domestic	No
9	Red	SUV	Imported	No
10	Red	Sports	Imported	Yes

b) ***Analyze*** following graph shows the output of a logistic regression model used to predict whether a patient is *infected* (Positive) or *not infected* (Negative) based on their RNA level. **[6]**

I. Find out the TP, TN, FP, FN. **[4]**

II. What is the impact of decreasing cut-off value? **[2]**



P.T.O

Q4. a) The table below shows the IMDb dataset of movies with their ratings, durations, and genres. *Apply* the K-Nearest Neighbors (KNN) classifier, classify the genre of the movie "Ted". Show all necessary steps. **[6]**

CLO 3

12 Marks

Rating	Movie Name	Duration	Genre
8.0	Mission Impossible	160	Action
6.2	Jason Bourne	170	Action
7.8	John Wick	165	Action
8.5	Avengers	180	Action
7.2	Dumb and Dumber	168	Comedy
8.2	Hangover	155	Comedy
6.8	Rush Hour	130	Comedy
7.0	The Mask	140	Comedy
7.6	The Pursuit of Happyness	117	Drama
8.3	Forrest Gump	142	Drama
6.9	The Notebook	124	Drama
8.1	A Beautiful Mind	135	Drama
7.4	Ted	170	?

b) *Solve* using the DBSCAN algorithm to the given data points and create the clusters with **minimum points = 2** and $\varepsilon = 3$. Use the Euclidean distance to calculate the distance between each points. Show clearly all necessary steps. Draw a graph containing core, border and noise points. **[6]**

A	B	C	D	E	F
1,2	2,2	4,5	7,8	9,3	25,8

Q5. Data have been downloaded from the Daraz 11-11 sale packages catalog by some business experts to learn the pattern of frequency of items. There are 9 different transactions (order 1 to 9) and each transactions have minimum 2 and maximum four items. Here 5 different kind of items are available. For simplicity Our Team lead used short names of items (I1- I 5) rather than full descriptive name (example I1= Hair dryer). Consider Minimum threshold support as 22.22% and confidence as 77.77%. *Apply* the most important analytics algorithm that is Apriori algorithm and find the frequent k items. Also indicate the association rules that are generated and highlight the strong ones, sort them by confidence. Show all your *steps* and *cut down transactions* with detail tabular format. **[12]**

CLO 3

12 Marks

Catalogue Item	List of Item in Package
Package 01	I1, I2, I5
Package 02	I2, I4
Package 03	I2, I3
Package 04	I1, I2, I4
Package 05	I1, I3
Package 06	I2, I3
Package 07	I1, I3
Package 08	I1, I2, I3, I5
Package 09	I1, I2, I3